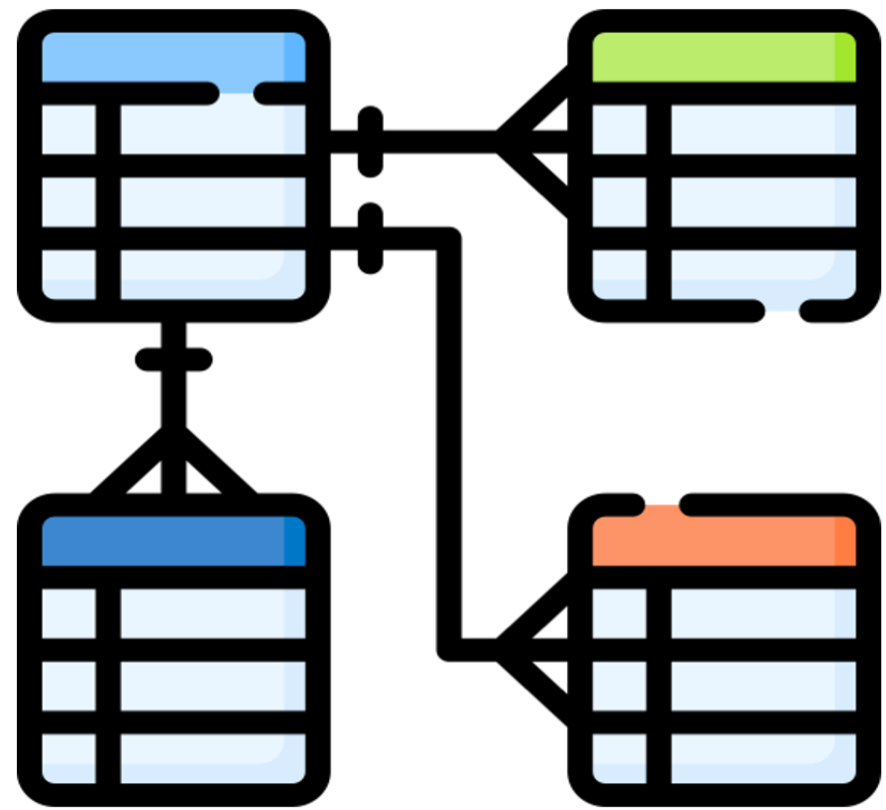




Project 1 & 2

Data Modelling
Python Coding
SQL



Source Dataset

- Flat schema about flight travel
- Format: CSV
- Total data size: 10.43 GB
- Total no. of files: 10
- Total queries: 20

```
departure_time  
arrival_time  
flight_id  
aircraft_id  
itinerary_no  
ticket_no  
flight_cost  
origin_airport  
destination_airport  
frequent_flier  
travel_date  
airplane_model  
frequent_flier_no  
passenger_name  
passenger_country  
tail_no  
distance  
turbulence  
temp_at_dept  
fuel_consumed_litre  
taxi_duration_mins  
avg_flight_speed_kmps  
engine_performance  
passenger_dob  
passenger_flight_class
```

Requirements



- The organization wants to capture some business insights from the travel data gathered over a period of time. Based on the insights, future decision would be taken to:
 - Add no. of flights
 - Optimize operations and cost
 - Float frequent flier deals
 - Optimize flight path
 - Figure out engine performance
 - Loss per mile etc.

Business Statements



- Average duration of flight.
- Which is the highest route that passengers use.
- Top 10 cities that have the highest take off and landing of flights.
- Top 10 flight paths that have high turbulence.
- If there any correlation between flight speed and distance.
- Which airplane model provides best fuel economy.
- Percentage of frequent flier year over year.
- How many flights were delayed.

Business Statements



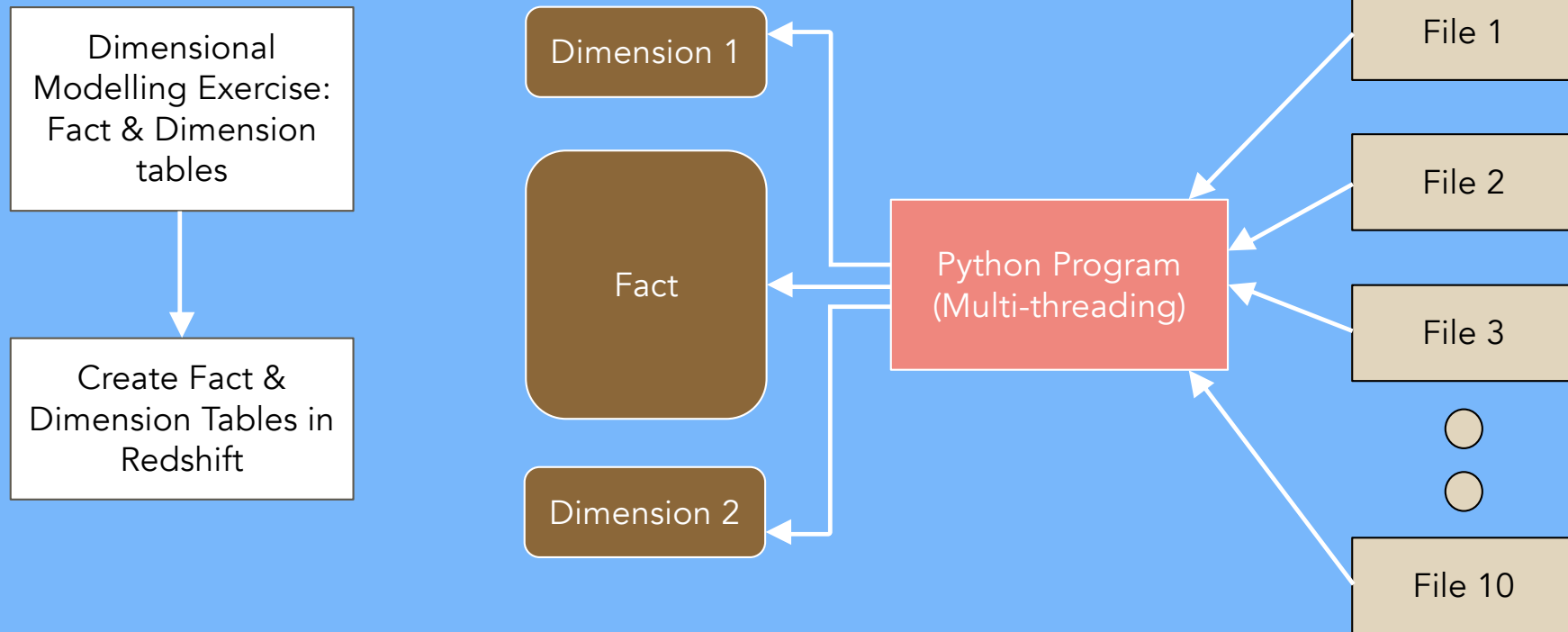
- Is there any correlation between delay of flight and the weather.
- When is engine performing the best. Performance is measured as average rpm per km. Lower rpm/km is better performance.
- How does the fuel consumption vary with distance of travel.
- Which country does the airline company get the most customers from.
- Which age group travels the most. Divide the age group into 4 as per your choice.
- Which age group uses which passenger class like "executive", "business", "economy" and "premium economy"

Business Statements



- Provide average passenger counts per flight, if possible.
- Need to find the average flight cost from source and destination. E.g. from a single source what is the average cost to all its destinations.
- Average fuel consumption in a month over the years.
- Which flight path (source and destination) has the highest turbulence.
- Which destination airport has the highest landing per year.
- Which is the busiest airport in the world over the years. Busiest airport is the one with highest takeoff and landing per year

Approach





Thank you

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